

On the Multiplicatively Closed Set Generated by 2 and 3

Partial results, an exact census up to 10^8 , and a rejected mean-field heuristic

Research dossier for Ben Green's open problem GREEN-036 (100 Open Problems, Number Theory) - August 25, 2026.

Source: <https://www.unsolvedmath.com/problems/GREEN-036> - status: **OPEN**

Abstract. Let A be the smallest set of positive integers containing 2 and 3 and closed under the operation $(a_1, a_2) \mapsto a_1 a_2 - 1$. Green's problem GREEN-036 asks whether A has positive asymptotic density. **The problem remains open.** This dossier contributes: (i) complete proofs of a well-foundedness structure theorem, of the modular obstruction (Lemma A: A avoids $1 \pmod 3$, whence upper density $d^*(A) \leq 2/3$), and of an explicit $\text{floor}((\text{floor}(\log_2 N) - 1)^{2/4}) \sim 0.5203 (\ln N)^2$ lower bound (Lemma B); (ii) an exact, independently cross-checked census $F(10^8) = |A \cap [1, 10^8]| = 55,939,931$ with per-decade statistics showing monotonically rising density $0.500 \rightarrow 0.559$ and local power-law exponents descending to 1.0079 ; (iii) a mean-field self-consistency heuristic $\sum_{b \in A} b^{-(1+\alpha)} = 1$ predicting $\alpha^* = 0.4734$, which is quantitatively rejected by the census (factor 9124), indicating that membership correlations, not pseudorandomness, govern A ; and (iv) a residue census showing that among all moduli $m \leq 500$ only $m = 3$ yields a proper closed residue set, so no improvement on $2/3$ is known. Statements are labeled [PROVEN], [COMPUTED], or [HEURISTIC] throughout.

1. Problem statement and status

Let A be minimal with $2, 3 \in A$ and $a_1, a_2 \in A$ implying $a_1 a_2 - 1 \in A$. Question: does the asymptotic density $d(A) = \lim F(N)/N$ exist and satisfy $d(A) > 0$, where $F(N) = |A \cap [1, N]|$? The question is **open**; nothing here settles it. Sections 2-5 establish partial results with proofs, Section 6 reports an exact computational census, and Section 7 derives a heuristic together with its own empirical rejection.

2. Well-foundedness, exact characterization, and generation algorithm

Structural fact [PROVEN]. 1 is not in A (seeds are 2, 3; $ab - 1 \geq 2 \cdot 2 - 1 = 3$ for $a, b \geq 2$), and whenever $n = a_1 a_2 - 1$ with $a_1, a_2 \in A$ one has $a_1, a_2 \geq 2$ and hence $a_1, a_2 \leq (n+1)/2 < n$. Values strictly decrease along any generation tree toward its leaves.

Theorem 1 (exact characterization) [PROVEN]. n is in A if and only if n is 2 or 3, or $n = ab - 1$ for some a, b in A . *Proof.* Minimality gives one direction directly: if $n > \max(a, b)$ equals $ab - 1$ with $a, b \in A$ then n cannot be excluded by any closed set containing the seeds, induction on $\max(a, b)$ being well-founded by the structural fact. Conversely the set $\{2, 3\} \cup \{ab - 1 : a, b \in A\}$ is contained in A by closure. ■

This makes A the least fixed point of $S \mapsto S \cup \{ab - 1\}$ and justifies the increasing-order dynamic program used by the verifier: membership bits in a bytearray; a min-heap of discovered-but-unexpanded values; a sorted partner list holding $A \cap [1, \sqrt{x}]$. Each popped minimum m is expanded against partners $a \leq \min(m, x/m) \leq \sqrt{x}$.

Theorem 2 (exactness) [PROVEN]. At termination the marked bytes are exactly $A \cap [1, X]$. *Proof.* Soundness is clear: inserted values $ma - 1$ use factors already certified to be in A . For completeness note pushes always exceed the popped value ($ma - 1 \geq 2m - 1 > m$), so pops occur in increasing order; by strong induction on n , before any pop of value $> n$ all elements of $A \cap [1, n]$ are discovered (a non-seed $n' \leq n$ has witness pair with larger factor $M \leq (n'+1)/2 < n$ already discovered and hence expanded while the heap minimum was still below n , inserting n'). Every pair (a, b) with product $\leq X$ is consumed at the expansion of its larger factor $M \leq (X+1)/2$, whose partner $\min(a, b) \leq \min(M, X/M) \leq \sqrt{X}$ is then present in the small list; so expansion against $A \cap [1, \sqrt{X}]$ misses nothing. ■

Cross-check [COMPUTED]. Independent naive fixpoint iteration reproduces the DP census exactly on $[1, 3000]$ ($|A| = 1358$ on both sides). Production run at $X = 10^8$: 6 min 31 s wall time, pure Python standard

library.

3. Lemma A: the modular obstruction and the 2/3 ceiling

Lemma A [PROVEN]. No element of A is congruent to 1 modulo 3. *Proof.* Work in $\mathbb{Z}/3$. The seed residues 2 and 0 both lie in $S = \{0, 2\}$. For r, s in S : if $r = 0$ or $s = 0$ then $rs - 1 = -1 = 2 \pmod{3}$; if $r = s = 2$ then $rs - 1 = 3 = 0 \pmod{3}$. Hence S is closed under the operation and contains the seeds, so by minimality the residues of A lie in S : never 1 modulo 3. ■

Corollary [PROVEN]. Upper density $d^*(A) \leq 2/3 = 0.666667$: each admissible class carries at most $N/3 + O(1)$ elements below N .

Census check [COMPUTED]. Zero of the 55,939,931 elements below 10^8 are 1 mod 3. Note A is a proper subset of the allowed classes: e.g. 6, 11, 62 are not in A although none is 1 mod 3 (their $n+1$ admit no divisor pair from $A \times A$).

4. Residue sets modulo arbitrary moduli

Proposition 3 [PROVEN]. For every modulus m the residue set $\{a \bmod m : a \text{ in } A\}$ equals exactly the minimal closed set S_m generated from $\{2 \bmod m, 3 \bmod m\}$. *Proof.* Containment: reduction commutes with the operation, so generation trees reduce into any closed set containing the seed residues. Reverse: every element of S_m arises after finitely many closure steps from the seeds, and each step lifts back through actual elements of A by induction on the step count. ■ Consequently $d^*(A) \leq |S_m|/m$ for every m , with CRT combination across coprime moduli.

modulus m	$ S_m $	$ S_m /m$	remark
3	2	0.666667	$S_3 = \{0, 2\}$: Lemma A
9	6	0.666667	$\{0, 2, 3, 5, 6, 8\}$: repackages mod 3
all other primes $p \leq 500$ (94)	p	1	$S_p = \mathbb{Z}/p$, full set
composite 4..500, m not in $\{9\}$	-	$> 2/3$	none beats $2/3$

Table 1. Closed residue sets [COMPUTED]: 95 primes and all composite moduli up to 500 scanned. Only $m = 3$ is proper; the best proven ceiling remains $2/3$ and the residue method is saturated at small heights.

5. Lemma B: an explicit logarithmic lower bound

Two families live in A [PROVEN]: $P_k = 2^k + 1$ ($k \geq 1$), the orbit of 3 under $x \mapsto 2x - 1$ (since $t - 1$ doubles along the orbit), and $V_j = (3^{j+1} + 1)/2$ ($j \geq 1$), the orbit of 2 under $x \mapsto 3x - 1$ (fixed point $1/2$: $t_k - 1/2 = 3^k(2 - 1/2)$). Closure produces all pairwise images.

Lemma B [PROVEN]. For $N \geq 8$, $F(N) \geq \text{floor}((\text{floor}(\log_2 N) - 1)^2 / 4) \geq (\log_2 N - 2)^2 / 4 = (\ln N)^2 / (4 \ln^2 2) + O(\ln N)$, with explicit constant $c = 1/(4 \ln^2 2) = 0.5203$.

Proof. Consider $z(i, j) = (2^{i+1} + 1)(2^{j+1} + 1) - 1$ for integers $0 \leq i \leq j$; all lie in A by two applications of closure, and $z = 2^{i+j+2} + 2^{i+1} + 2^{j+1}$. **Distinctness.** If $i < j$ then $z = 2^{i+1}(1 + 2^{j-i} + 2^{j+1})$ with odd bracket, so $v_2(z) = i+1$ and odd part $q = 1 + 2^{j-i} + 2^{j+1}$. If $i = j \geq 1$ then $z = 2^{i+2}(2^{i+1} + 1)$: $v_2(z) = i+2$, odd part $2^i + 1$. The degenerate pair $(0, 0)$ gives $z = 8 = 2^3$, the only pure power of two in the family (every off-diagonal entry has odd part $q \geq 11$ and value ≥ 14 ; every diagonal entry with $i \geq 1$ has odd part $2^i + 1 > 1$ and value ≥ 24). Suppose $z(i, j) = z(i', j')$. Off-diagonal vs off-diagonal: equal v_2 forces $i = i'$, and equality of odd parts reads $2^{j-i}(1 + 2^{j-i}) = 2^{j'-i}(1 + 2^{j'-i})$; cancelling the odd integer forces $j = j'$. Diagonal vs diagonal: equal v_2 forces $i = i'$. Off-diagonal vs diagonal would force simultaneously $i+1 = i'+2$ and $1 + 2^{i+1} + 2^{i+2} = 2^{i'+1} + 1$, i.e. $2^{i+1}(1 + 2^{i+1}) = 2^{i'+1}$ with $d \geq 1$: impossible since a pure power of two cannot be divisible by the odd integer $1 + 2^{i+1} \geq 3$. Hence the map is injective. **Counting.** Let $\lambda = \text{floor}(\log_2 N)$; whenever $s := i+j+3 \leq \lambda$, $z \leq 2^{s-1} + 2^{s-2} + 2^{s-2} = 2^s$ with strict inequality (equality would force $s = 2(s-2)+3$), so $z < 2^s \leq N$. The pairs with $i \leq j$, $i+j \leq \lambda-3$ number $\sum_{s=0}^{\lambda-3} \text{floor}(s/2)+1 = \text{floor}((\lambda-1)^2/4)$. ■

Mixed family [COMPUTED; injectivity unproven]. With $U_i = 2^{i+1} + 1$ and $V_j = (3^{j+1} + 1)/2$, the values $w(i, j) = U_i V_j - 1$ all lie in A ; a full injectivity proof would raise the constant to $1/(2 \ln 2 \ln 3) = 0.6566 (\ln N)^2$.

The 2-adic cascade recovers $\min(i+1, v_2(V_j - 1))$ but ties block complete recovery. Computed up to 10^8 : 216 values, zero collisions, all in A; the proven family contributes 168 values against the proven floor 156. Both bounds are exponentially far from the true $F(10^8) = 55,939,931$: their value is unconditional methodology, not approximation.

6. Exact census up to 10^8

k	$F(10^k)$	$F(10^k)/10^k$	local exp	harm incr	delta_hat
1	5	0.500000	-	1.2694	0.5513
2	39	0.390000	0.8921	0.8178	0.3552
3	422	0.422000	1.0342	0.9470	0.4113
4	4,805	0.480500	1.0564	1.1029	0.4790
5	51,508	0.515080	1.0302	1.1838	0.5141
6	535,585	0.535585	1.0170	1.2318	0.5350
7	5,493,428	0.549343	1.0110	1.2637	0.5488
8	55,939,931	0.559399	1.0079	1.2871	0.5590

Table 2. Exact per-decade census [COMPUTED]. local exp = $\log(F(10^k)/F(10^{k-1}))/\log 10$; harm incr = sum of $1/b$ over $A \cap (10^{k-1}, 10^k]$; delta_hat = incr / $\ln 10$ estimates density.

Reading of the data [COMPUTED trend; OPEN conclusion]: raw density rises in every decade; local exponents descend toward 1 from above (top value 1.0079); per-decade harmonic mass approaches delta_hat $\ln 10$ with delta_hat climbing through 0.56. A log-log least-squares fit over decades 4-8 gives $\alpha_{OLS} = 1.01600 \pm 0.00254$ with $R^2 = 0.99998$, consistent with near-linear growth $F(N) \sim c_N N$ and drifting density. All trends are compatible with convergence of $F(N)/N$ to a positive limit $\leq 2/3$; neither existence nor positivity of the limit is provable by these means.

Additional exact facts within the completely stored range $[1, 10^6]$ ($F = 535,585$) [COMPUTED]: first elements 2, 3, 5, 8, 9, 14, 15, 17, 23, 24, 26, ...; smallest non-members 1, 4, 6, 7, 10, 11, 12, 13, 16, 18, ...; maximal gap between consecutive elements is 10 (2159 to 2169); conditional density inside allowed classes $\{n \not\equiv 1 \pmod 3\}$ rises monotonically: 0.591, 0.634, 0.721, 0.773, 0.803 at $N = 10^2, \dots, 10^6$.

7. Mean-field heuristic: derivation and empirical rejection

Membership is an OR over divisor pairs (Theorem 1): n is in A iff $n+1 = ab$ for some a, b in A (or n in $\{2, 3\}$). Assume **(H1)** equidistribution: for each b , $|A \cap [1, M] \cap (-1 \pmod b)| \sim F(M)/b$ uniformly in b . Counting candidate images $ba - 1 \leq N$ over all b in A gives $F(N) \sim \sum_{b \in A} F(N/b)/b$. Inserting the ansatz $F(N) \sim c_N N^\alpha$ and cancelling yields the self-consistency equation (*): $\sum_{b \in A} b^{-(1+\alpha)} = 1$. Because the truncated harmonic sum diverges like delta_hat $\ln X$ (computed $H_X = 9.1036$ at $X = 10^8$, increments approaching 1.287 per decade), the left side decreases continuously from infinity to zero, so under (H1) a unique $\alpha^* > 0$ exists [HEURISTIC].

Numerics [COMPUTED]. Exact truncated sums below $2 \cdot 10^5$ and 64 logarithmic buckets per decade above give $\alpha^*_{trunc}(10^8) = 0.47344208837849144$ (root residual $+2.5e-11$; stable: 0.472709 already at $X = 10^6$). The integral surrogate $\delta^{2-\alpha} / \alpha = 1$ gives $\alpha^*(\delta = 0.5594) = 0.4185$ and $\alpha^*(\delta = 2/3) = 0.4785$.

Rejection [COMPUTED vs HEURISTIC]. The mean field predicts $F(10^8) \sim (10^8)^{0.4734} = 6,131$ against the exact 55,939,931: off by factor 9,124. Simultaneously the direct fit sits at $\alpha_{OLS} = 1.016 > 1 > \alpha^*$. Two mechanisms identify the failure: (1) *self-affine imaging* - each b in A transports a near-copy $\{ba - 1 : a \in A\}$ into A , so membership correlations inflate density beyond products of marginals (the OR-condition over divisor pairs); (2) *modular rigidity* - equidistribution fails absolutely on class $1 \pmod 3$, and Proposition 3 formalizes such constraints for every modulus. Equation (*) therefore describes a random model of A , not A itself. Its interest is negative but sharp: GREEN-036 is not governed by pseudorandom multiplicative statistics, and any successful approach must exploit the correlated, self-affine structure.

8. Outlook: what is missing

[OPEN] To prove $d(A) > 0$ one wants $F(N) \geq cN$, plausibly via eventual containment of an arithmetic progression of allowed class, or a bootstrap converting lower density at height T into higher density at height $T' > T$. Visible obstructions: 6, 11, 62 are absent despite allowed residues and the gap 2159-2169 shows early holes, so any progression claim needs a threshold phenomenon. To prove $d(A) = 0$ one must reverse a trend rising across eight decades of exact computation; no mechanism suggests itself. Between the two: the only proven ceiling is $2/3$ (Lemma A), the residue method is exhausted for $m \leq 500$, and the proven growth floor is $(\ln N)^2/(4 \ln^2 2)$. Any genuinely new constraint - larger-modulus rigidity, p-adic restrictions, complement stability - would constitute real progress on GREEN-036.

9. Attack iteration: obstructions and rigidity

L1 (image counting cap). $T_b(x) = bx - 1$ is injective, so $|T_b(A) \cap [1, N]| = F(\text{floor}((N+1)/b))$ exactly; and $|T_{b_1} A \cap T_{b_2} A \cap [1, N]| \leq \text{floor}((N+1)/\text{lcm}(b_1, b_2))$. The pairwise overlap load over seeds $b \leq B$ satisfies $S(B) = \sum \gcd(b_1, b_2)/(b_1 b_2) = (1/\pi^2)(\log B)^3 + O((\log B)^2)$ (phi-identity), so Bonferroni's positive part collapses once $(\log B)^3 / \pi^2 \gtrsim \delta \log B$: images of a dense low-scale region cannot certify density growth. Finite evidence at $X = 2e6$: $|(3A-1) \cap (5A-1)| = 132,814 \leq 133,333$ (99.6% tight).

L2 (multiplication-table cap; no one-step bootstrap). By Erdos 1960 (quantified by Ford, Annals 2008): $M(N) = |\{ab: a, b \leq N\}| = N^2/(\log N)^{\delta_F} (\log \log N)^{3/2}$, $\delta_F = 1 - (1 + \log \log 2)/\log 2 = 0.08607$. Restricting to any $S \times S$, $S \subset [2, N]$, only shrinks the table and $ab \rightarrow ab - 1$ is a value bijection; hence one closure step on ANY dense seed yields relative density $o(1)$ at height N^2 . No implication $F(N) \geq \delta N \Rightarrow F(N^2) \geq \Phi(\delta) N^2$ follows from the closure rule alone: density at height N^2 must be built through $\sim \log N$ intermediate scales.

L3 (complement propagation along primes). If p is prime and $p \notin A$, then $pq - 1 \notin A$ for every prime q (semiprime pq has no other factor pairs). Anchor: $11 \notin A$ since $12 = 2 \cdot 6 = 3 \cdot 4$ with 4, 6 out. All 9,592 numbers $11q - 1$ (q prime $\leq 1e5$) are outside A (finite evidence). Such families count $O(N/\log N)$: propagation complements have density 0 and cannot obstruct $d(A) > 0$.

L4 (2-chain test; prime rigidity). For $m \geq 2$: $2m - 1$ in A iff m in A or some other divisor pair (a, b) of $2m$ lies in $A \times A$. In particular for primes $p \geq 5$: p in $A \iff qp - 1$ in A for every prime $q \neq p$.

L5 (powers of two; finite evidence). Exact memoized recursion gives, for $k \leq 40$: 2^k in A iff k in $\{1, 3, 9, 15, 17, 21, 23, 25, 27, 29, 33, 35, 39\}$. No even $k > 0$ occurs; odd k cascade through $(2^{k+1})/3$ with unbounded certificate depth - e.g. 512 in A via $513 = 3 \cdot 171$, $172 = 2 \cdot 86$, $87 = 3 \cdot 29$, $30 = 2 \cdot 15$, $16 = 2 \cdot 8$, $9 = 3 \cdot 3$ (depth ≥ 5). No bounded-depth local rule captures A .

The GAP. Nothing lower-bounds the count of $n \leq X$ for which $n+1$ has a divisor pair in $A \times A$ beyond the $O((\log X)^2)$ families of Lemma B; positive density requires at least cX such n . Status after this iteration: STILL OPEN, with sharpened obstruction theory.

10. Attack II: progression holes, a doubling recurrence, rescue

T1 [PROVEN]. For every reduced class $r \bmod M$ ($\gcd(r+1, M)=1$): infinitely many $n \equiv r \pmod{M}$ with $n \notin A$; count $\Omega_M(X/\log X)$. Proof: fix prime $p \equiv 1 \pmod{3M}$ (Dirichlet), so $p \notin A$ by Lemma A; primes $q \equiv r+1 \pmod{M}$ are infinite (Dirichlet); $pq-1$ is semiprime-blocked by the divisor-pair characterization. Consequence: A contains no infinite arithmetic progression - eventual-progression routes to positive density die.

T2 [PROVEN, density-vacuous]. $4q-1 \notin A$ for prime $q \equiv 5 \pmod{6}$ ($4 \notin A$ blocks two pairs; $2q \equiv 1 \pmod{3}$ blocks the rest) - but every such n lies in the dead class $1 \pmod{3}$. Elementary blocking lands in the dead class or gets rescued: $116 = 9 \cdot 13 - 1$ in A via pair $(3, 39)$; the naive three-prime block fails since $364 = 7 \cdot 13 \cdot 4 = 14 \cdot 26$ with $14, 26$ in A gives 363 in A .

T3 [PROVEN] doubling recurrence. $F(2N) \geq F(N) + [F(N) - F(N/2)] + [F(2N/3) - F(N/3)] - \text{floor}(2N/6) + \text{floor}(N/6) - 2$: windows $W=\{2a-1\}$, $V=\{3a-1\}$ confined to $(N, 2N]$, lcm-overlap only on multiples of 6, prefix overlap at most the point N . Relative form: $\delta(2N) \geq 13 \delta/12 - 1/8 - O(1/N)$. Slope $13/12 > 1$ with fixed point $-3/2$: valid at every scale (computed margins $+262$ to $+261,658$) but CONTRACTING - the linear overlap penalty compounds against the polylog seed. The gap is structural.

11. Attack III: method barrier and conditional growth

Theorem 3.1 [PROVEN] method barrier. $C = \{n : n \not\equiv 1 \pmod{3}\}$ contains $\{2,3\}$, is closed under $ab-1$ ($0*s-1\equiv 2, 2*2-1\equiv 0$), and has density exactly $2/3$; A is the minimal closed set. Hence closure plus Lemma A brackets the answer between log-growth and $2N/3$: any resolution must invoke arithmetic strictly beyond these two axioms.

Lemma [PROVEN] comb evaluation. Left-combs $v_1=l_1, v_k = v_{\{k-1\}} l_k - 1$ over l_j in A lie in A with exact expansion $v_r = \text{prod}(l) - \sum_{\{k \geq 3\}} \text{prod}_{\{j \geq k\}}(l_j) - 1$ (2000/2000 checks; 3351/3351 combs $\leq 1e6$ in census).

Theorem 3.2 [FALSIFIED] multi-orbit injectivity. The fold map $(i,j,k) \rightarrow (U_i V_j - 1)W_k - 1$ ($U=2^{k+1}$ family, $V=(3^{j+1}+1)/2, W=(7*5^{k+1})/4$) has 5 collision certificates $\leq 1e6$: 395, 9845, 49220, 48135, 240635 - caused by orbit coincidences $W_1=U_2=9$ and $V_6=W_4=1094$ and the S-unit coincidence $3*365=5*219$. All 346 folds ARE members. Raw ternary products $UVW-1$ are NOT members (6563 witness).

Proposition 3.3 [OBSTRUCTION/BARRIER]. Modulus scan: primes ≤ 1500 ; prime powers $2^k \leq 4096, 3^k \leq 6561$; semiprimes exhaustive ≤ 1900 plus 190 moduli ≤ 5000 : the ONLY proper closed residue systems are 3-primary lifts $S_m = \pi_3^{-1}(\{0,2\})$ with ratio exactly $2/3$ (CRT-lift lemma). Lemma A is the complete congruential obstruction; Target D needs non-congruential input.

Theorem 3.4 [REDUCED TO CONJECTURE]. Cross-scale richness $C1'(\beta)$: $|\text{union}_{\{b \in A_{\{2M\}}\}} A_M \cap b(A_M) \cap [1, 2M^2]| \geq c M^{1+\beta}$. Under $C1'$ at infinitely many scales, $F(N) \geq N^\alpha$ along a subsequence with $\alpha=(1+\beta)/2$; all-scales self-refined variant gives $F(N) \geq N^{1-o(1)}$. Implication chain proved; conjecture open.

Theorem 3.5 [OBSTRUCTION/BARRIER]. Same-scale tables are GP-thin: $|A_M * A_M \cap [1, 2M]| = 881$ vs $F=422$ at $M=1e3$; 9851 vs 4805 at $1e4$ - ratios 2.09/2.05 vs geometric-progression floor ~ 2 . All single-scale arguments (images or tables) are excluded; multiscale schemes need $C1'$ -type input.

Reproducibility. Everything regenerates by python3 verify_solution.py [X] (default $X = 10^7$, about 20 s; production $X = 10^8$, 6.5 min, nine of nine checks passed). Artifacts: data/census_summary.json, data/census.csv.gz (first 10^6 elements), data/run_output.txt reproduced verbatim in the Appendix; targeted checks regenerate by attack_verify.py, attack2_verify.py and attack3_verify.py (outputs also in the Appendix). Standard library only; deterministic.

The complete dossier is version-controlled at <https://github.com/GamebP/MathProblems/tree/main/GREEN-036>, mirroring this folder file-for-file.

Appendix. Verbatim verifier output (production run, $X = 10^8$)

GREEN-036 verification run, $X = 100000000$

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[1] exact census DP: |A cap [1,100000000]| = 55939931 (351.06s)
[2] naive closure cross-check at X=3000: DP |A|=1358, naive |A|=1358, identical=True (0.57s)
[3] Lemma A: elements congruent to 1 (mod 3) in census: 0 (out of 55939931); predicted 0
    observed raw density  $F(100000000)/100000000 = 0.559399$  (proven upper bound  $2/3 = 0.666667$ )
[4] count table |A cap [1,10^k]| and per-decade increments (25.78s):
      k  F(10^k)  increment  F/N  local exp  harm incr  delta_hat
      1      5      5      0.500000      nan      1.2694  0.5513
      2     39     34      0.390000      0.8921  0.8178  0.3552
      3    422    383      0.422000      1.0342  0.9470  0.4113
      4   4805   4383      0.480500      1.0564  1.1029  0.4790
      5  51508  46703      0.515080      1.0302  1.1838  0.5141
      6 535585 484077      0.535585      1.0170  1.2318  0.5350
      7 5493428 4957843      0.549343      1.0110  1.2637  0.5488
      8 55939931 50446503      0.559399      1.0079  1.2871  0.5590
[5] Lemma B lower-bound families:
    P-orbit  $2^k+1$ ,  $k=1..26$ : 26 values, all in A: True
    Z-products  $(2^{i+1})(2^{j+1})-1 \leq X$  ( $i \leq j$ ): generated 168, in A: 168, collisions: 0
    proven counting formula  $\text{floor}((\text{floor}(\log_2 X)-1)^2/4) = 156$  ( $\leq$  actual 168)
    lower bound achieved:  $168 = 3.00322e-06 \times F(100000000)$ 
    mixed  $(2^{i+1})(3^{j+1}/2)-1 \leq X$ : generated 216, in A: 216, collisions: 0
    asymptotic constants: pure-2 (PROVEN)  $1/(4 \ln^2 2) = 0.52034$ , mixed (if injective)  $1/(2 \ln 2 \ln 3) = 0.65660$ 
    (0.00s)
[6] residue census, 95 primes  $\leq 500$  (6.46s):
    primes with PROPER  $S_p$ : [(3, 2)]
    composite moduli 4..500 beating 2/3: none
     $S_3 = [0, 2]$  (Lemma A),  $|S_p| = p$  for all other primes  $p \leq 500$ : True
    combined CRT bound over proper prime constraints: 0.666667 (= proven 2/3 here); best single 0.666667
[7] growth-exponent estimates:
    OLS log-log fit over  $k=4..8$ :  $\alpha_{\text{hat}} = 1.01600$  ( $R^2 = 0.999981$ )
    top local exponent  $\ln(F_8/F_7)/\ln 10 = 1.00788$ 
    truncated harmonic sum  $H_X = \sum_{b \in A \text{ cap } [1,100000000]} 1/b = 9.103598$ 
    truncated zeta grid:  $S(0.0)=9.1039$ ,  $S(0.25)=1.8511$ ,  $S(0.5)=0.9483$ ,  $S(0.75)=0.6321$ ,  $S(1.0)=0.4615$ 
    solved  $S_X(\alpha^*) = 1$ :  $\alpha^* \cdot \text{trunc} = 0.473442$  (residual +2.51e-11)
    mean-field prediction  $F(100000000) \sim X^{\alpha^*} = 6131$  vs actual 55939931 (ratio 9124)
[8] artifacts written in 0.9s: census_summary.json, census.csv.gz (first 1000000 elements) sizes: 3618B, 2285592B
=====
[PASS] EXACTNESS      DP set == naive fixpoint closure on [1,3000] (|A| = 1358 both sides: True)
[PASS] LEMMA-A        0 violations of  $n \equiv 1 \pmod 3$  among 55939931 census elements
[PASS] PORBIT         all 26 values  $2^k+1 \leq X$  lie in A
[PASS] LEMMA-B-Z      168/168 Z-products in A, 0 collisions, proven formula 156  $\leq$  actual
[PASS] LEMMA-B-MIXED  216/216 mixed products in A, 0 collisions (injectivity COMPUTED, not proven)
[PASS] RESIDUES        $S_3 = \{0,2\}$ ; proper-prime list [(3, 2)]; combined bound 0.666667  $\geq$  observed density 0.559399
[PASS] MONOTONE       decade counts strictly increasing, e.g. 5 -> 55939931
[PASS] ZETA-ROOT       $S_X(\alpha^*) = 1$  at  $\alpha^* = 0.473442$ , residual +2.51e-11
[PASS] DENSITY-TREND  F/N still rising: 0.5493 -> 0.5594 (question OPEN)
=====
9/9 checks passed. Problem GREEN-036 remains OPEN; this run establishes computational evidence only.
```

Appendix B. Verbatim attack-check output (attack_verify.py, $X = 2e6$)

GREEN-036 attack checks – every output below is FINITE EVIDENCE

```
=====
[0] census rebuilt to  $X=2000000$ :  $F(X)=1080418$ 
(i)  $2^k$  in A for  $k \leq 40$ :  $k$  in [1, 3, 9, 15, 17, 21, 23, 25, 27, 29, 33, 35, 39] (FINITE EVIDENCE,  $k$  range 1..40)
(ii) 11 in A: False
(ii) holes  $11q-1$  ( $q$  prime  $\leq 1e5$ ): 9592 tested, 0 members of A (expect 0) (FINITE EVIDENCE)
(iii)  $b=(2,3)$   $\text{lcm}=6$ :  $|\text{intersection}|=324421 \leq \text{floor}((X+1)/\text{lcm})=333333$  [OK] (FINITE EVIDENCE)
(iii)  $b=(2,5)$   $\text{lcm}=10$ :  $|\text{intersection}|=132779 \leq \text{floor}((X+1)/\text{lcm})=200000$  [OK] (FINITE EVIDENCE)
(iii)  $b=(3,5)$   $\text{lcm}=15$ :  $|\text{intersection}|=132814 \leq \text{floor}((X+1)/\text{lcm})=133333$  [OK] (FINITE EVIDENCE)
(iii)  $b=(8,9)$   $\text{lcm}=72$ :  $|\text{intersection}|=27675 \leq \text{floor}((X+1)/\text{lcm})=27777$  [OK] (FINITE EVIDENCE)
(iv) 19 in A: False (expect False) (FINITE EVIDENCE)
(iv) 25 in A: False (expect False) (FINITE EVIDENCE)
(iv) 205 in A: False (expect False) (FINITE EVIDENCE)
(iv) 1024 in A: False (expect False) (FINITE EVIDENCE)
(iv) support facts: 41 in A: True; 14 in A: True; 13 mod 3 = 1; 103 mod 3 = 1
=====
done
```

Appendix C. Verbatim output (attack2_output.txt)

GREEN-036 attack-2 checks – FINITE EVIDENCE unless stated PROVEN

```
=====
[T2]  $4q-1$ ,  $q \equiv 5 \pmod 6$ ,  $q < 20000$ : 1136 tested, 0 in A (PROVEN expect 0); all  $n \equiv 1 \pmod 3$ : True
[T1]  $M=5, r=2$ :  $p=31$  (prime  $\equiv 1 \pmod{15} \Rightarrow$  PROVEN not in A),  $q$  prime  $\equiv 3 \pmod 5$  ( $< 6e4$ ): 1516 values  $pq-1$ , 1516 confirmed
    holes, 0 members (PROVEN expect 0).
[T1] first holes: [92, 402, 712, 1332, 1642, 2262, 2572, 3192]
[R] 116 in A: True (rescued via pair (3,39)); 39 in A: True; 13 in A: False
[R] refuted-conjecture witness  $364=7 \cdot 13 \cdot 4=14 \cdot 26$ : sides 14,26 in A: True,True; 363 in A: True
[T3] doubling recurrence valid at  $N=1e3..1e6$ : True, margins [(1000, 262), (10000, 2618), (100000, 26083), (1000000,
    261658)]
[T3] relative form  $\text{delta}(2N) \geq 13\text{delta}/12 - 1/8 - O(1/N)$  at  $1e5, 1e6$ : True
[A] Lemma A violations in census to 2000000: 0
=====
done
```

Appendix D. Verbatim output (attack3_output.txt)

GREEN-036 attack-3 checks – FINITE EVIDENCE unless stated PROVEN

```
=====
[3A] comb identity 2000/2000: True; combs<=1e6 in A: 3351/3351
[3B] C={0,2} mod 3 closed under rs-1: True [PROVEN separately]
[3.2] orbit coincidences: W_1=9==U_2=9: True; V_6=1094==W_4=1094: True; 3*365==5*219: True
[3.2][FALSIFIED] ternary 6563 witness: 3*2*1094-1=6563, member=False
[3.2] depth-3 folds <=1e6: 346 generated, 341 distinct, 5 collisions, not-in-A 0
      collision: (((0, 5, 1), True), (1, 0, 4), 9845, True)
      collision: (((1, 0, 5), True), (1, 6, 1), 49220, True)
      collision: (((1, 0, 2), True), (2, 1, 1), 395, True)
      collision: (((0, 5, 2), True), (2, 1, 4), 48135, True)
      collision: (((1, 6, 2), True), (2, 1, 5), 240635, True)
[3.3] primes<=1500: proper = [(3, 2)] (expect [(3,2)])
[3.3] prime powers: [('m', '6/9'), ('m', '18/27'), ('m', '54/81'), ('m', '162/243'), ('m', '486/729'), ('m',
'1458/2187'), ('m', '4374/6561')]
[3.3] semiprimes<=1900 exhaustive (91 moduli): non-3-lift proper = []
[3.3][extended] 190 semiprime moduli <=5000 exhaustive (p<q<=71): non-3-lift proper = []
[3.5] M=1000: F(M)=422, |SS cap[1,2M]|=881, ratio=2.088 (GP floor ~2)
[3.5] M=10000: F(M)=4805, |SS cap[1,2M]|=9851, ratio=2.050 (GP floor ~2)
[A] Lemma A violations <= 1000000: 0
=====
done
```